# Project Report

**Project Name: Microcontroller Based Grid Voltage Surge and Current Protector device.**

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**N.B: Please open** *...riad2008004/frontier_vs_oc/projects_files/circuit_diagram.pdf and ...riad2008004/frontier_vs_oc/projects_files/annotated_circuit_diagram* **for easy understanding.**

**Design Approch:** This project includes Arduino Uno as the main microcontroller. Input sections include ACS712 30A current sensor with 10nF filtering capacitor (Medium Filtering), LM258 based AC voltage step down circuit with +5V positive bias (to keep the output positive). 5 push buttons added to control the menu. As the output devices JHD 16*2 I2C LCD Display is used, as well as serial output is added to show the output in computer. To connect and disconnect the load one relay is connected with a fault detection RED LED. Three 8.2A Motor is used as the load to perform the simulation. A diode with the Load is connected in forward bias to operate the DC motors with lower complexity. Three switches with the load is connected to control the motors to prove the steps of current consumption. A switch with the voltage step down circuit is added to test voltage surge workability.

**Features:**

**AC overcurrent and voltage surge protection:** The protector device runs continuously when the current and voltage both values is under predefined voltage and ampere. When the load current or voltage or both exceed the safe value the relay trigged with Fault LED ON and is isolates the load from the grid.

**Fault Reset:** When the device go in fault condition, the display shows the fault type whether it is OC or OV or both and a text is shown "*Hold btn 5s reset*". At that time if any menu buttons of the device is pressed for 5s it will reset the device, and the device will run as it was.

**Settings update**: When the device in running condition, if any buttons of the device is pressed for 5s that will go to settings mood. It will show "*Voltage limit \n V L-/R+ SET"* and "*Current limit \n C L-/R+ SET"*. Up and Down buttons can be used to alter the settings menu with Current limit to Voltage limit and vice versa. Right and Left buttons are used to change the values of the settings. Pressing SET will save the values to the device and go to its operation.

**Limitations:** The settings value is only saved by variable change, it will change when the power cut-of. Use of EEPROM can overcome this. Here it cannot be implemented due to proteus limitations. The current sensing, and voltage sensing circuits can be replaced by Transformer type circuits for better output in real life operations. The device settings can not be changed through computer. It can be implemented that the device will read the serial data from computer and change its internal value if necessary.